Exercises ISLR – Ch.13

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Conceptual

Exercises: 2, 4

Ex.2) Rejection random variable (a)

2. Suppose that we test m hypotheses, and control the Type I error for each hypothesis at level α. Assume that all m p-values are independent, and that all null hypotheses are true.

   (a) Let the random variable A_j equal 1 if the jth null hypothesis is rejected, and 0 otherwise. What is the distribution of A_j?

Since all null hypotheses are true, their individual probability of rejection is the same.

Therefore, each random variable follows a **Bernoulli distribution,** whose parameter is the Type I error level.

$$P(A_j) = \begin{cases} P(A_j = 1) = \alpha \\ P(A_j = 0) = 1 - \alpha \end{cases} \quad \Longrightarrow \quad \boldsymbol{A_j \sim Ber(\alpha)}$$

Ex.2) Rejection as a random variable (b,c)

(b) What is the distribution of $\sum_{j=1}^{m} A_j$?

Since the random variables are independent and identically distributed (iid), the sum of Bernoulli variables follows a **Binomial distribution.**

$$\sum_{j=1}^{m} \boldsymbol{A_j} \sim \boldsymbol{Bin}(\boldsymbol{m}, \boldsymbol{\alpha}) \quad \Rightarrow \mathrm{P}\left(\sum_{j=1}^{m} A_j = x\right) = \binom{m}{x} \alpha^x (1-\alpha)^{m-x}$$

(c) What is the standard deviation of the number of Type I errors that we will make?

Since the number of Type I errors follow a **Binomial distribution**, the standard deviation can be calculated leveraging the properties of this distribution.

$$Var\left(\sum_{j=1}^{m} A_j\right) = m\alpha(1-\alpha) \qquad \Rightarrow \qquad \boldsymbol{SD}\left(\sum_{j=1}^{m} \boldsymbol{A_j}\right) = \sqrt{\boldsymbol{m\alpha(1-\alpha)}}$$

Ex.4) FWER and FDR (a)

4. Suppose we test $m = 10$ hypotheses, and obtain the p-values shown in Table 13.4.

(a) Suppose that we wish to control the Type I error for each null hypothesis at level $\alpha = 0.05$. Which null hypotheses will we reject?

Null Hypothesis	p-value	
H_{01}	0.0011	✖
H_{02}	0.031	✖
H_{03}	0.017	✖
H_{04}	0.32	
H_{05}	0.11	
H_{06}	0.90	
H_{07}	0.07	
H_{08}	0.006	✖
H_{09}	0.004	✖
H_{10}	0.0009	✖

TABLE 13.4. *p-values for Exercise 4.*

6 rejections:
- H01
- H02
- H03
- H08
- H09
- H10

Ex.4) FWER and FDR (b)

(b) Now suppose that we wish to control the FWER at level $\alpha = 0.05$. Which null hypotheses will we reject? Justify your answer.

Bonferroni method: equivalent individual cutoff → **0.05/10 = 0.005**

Null Hypothesis	p-value	
H_{01}	0.0011	✖
H_{02}	0.031	
H_{03}	0.017	
H_{04}	0.32	
H_{05}	0.11	
H_{06}	0.90	
H_{07}	0.07	
H_{08}	0.006	
H_{09}	0.004	✖
H_{10}	0.0009	✖

TABLE 13.4. *p-values for Exercise 4.*

3 rejections:

- H01
- H09
- H10

Bonferroni: **conservative**

Ex.4) FWER and FDR (c, d)

(c) Now suppose that we wish to control the FDR at level $q = 0.05$. Which null hypotheses will we reject? Justify your answer.

(d) Now suppose that we wish to control the FDR at level $q = 0.2$. Which null hypotheses will we reject? Justify your answer.

Benjamini-Hochberg method:

i) Order p-values

ii) Increase cutoff linearly q/m

H0	Ordered p-value	Cutoff q=0.05	Cutoff q=0.2
H10	0.0009	0.005 ✗	0.02 ✗
H01	0.0011	0.010 ✗	0.04 ✗
H09	0.004	0.015 ✗	0.06 ✗
H08	0.006	0.020 ✗	0.08 ✗
H03	0.017	0.025 ✗	0.10 ✗
H02	**0.031**	**0.030**	0.12 ✗
H07	0.07	0.035	0.14 ✗
H05	0.11	0.040	0.16 ✗
H04	**0.32**	0.045	**0.18**
H06	0.90	0.05	0.20

5 rejections: q=0.05

- H10
- H01
- H09
- H08
- H03

8 rejections: q=0.2

- H10
- H01
- H09
- H08
- H03
- **H02**
- **H07**
- **H05**

Ex.4) FWER and FDR (e)

(e) Of the null hypotheses rejected at FDR level $q = 0.2$, approximately how many are false positives? Justify your answer.

The expected number of false positives is related to the total number of positives (rejected hypotheses) and the cutoff (q).

$$FDR = E\left[\frac{False\ Positives}{Total\ Positives}\right] \approx q = 0.2$$

Using this approximation for the number of rejected hypotheses (8):

$$E\left[\frac{False\ Positives}{8}\right] = 0.2$$

$$E[False\ Positives] = \mathbf{1.6 \approx 2}$$